Lab 8 Assignment: Automating Multi-Container Deployment with CI/CD

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Task 1: Add Database + Docker Compose

Database Choice

I chose **MongoDB** as the database.

Reason for using service name as hostname:
In Docker Compose, the app container uses the database service name (mongo) as the hostname instead of localhost or an IP because Docker Compose provides automatic DNS resolution for services. This ensures the app can always connect to the database regardless of the container's dynamic IP.

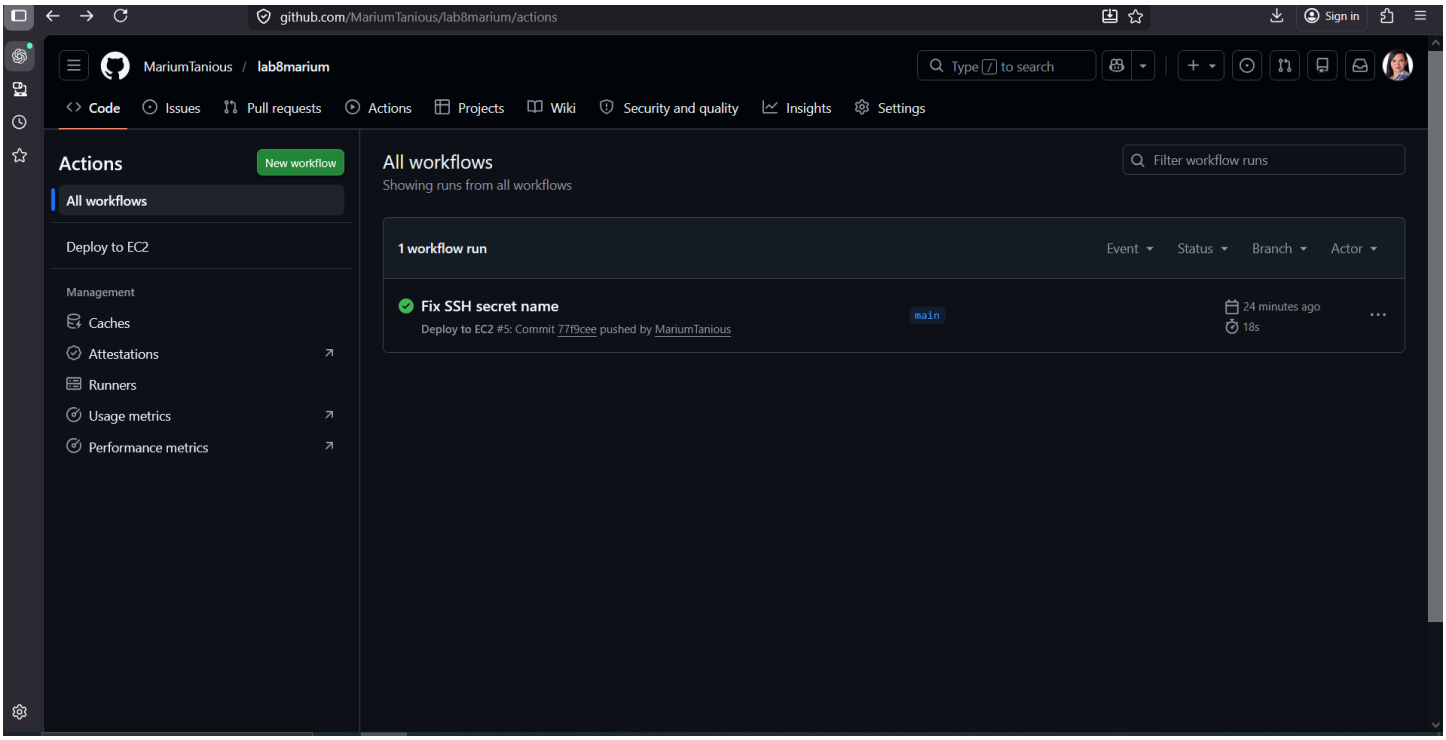
1.Docker Compose Status:

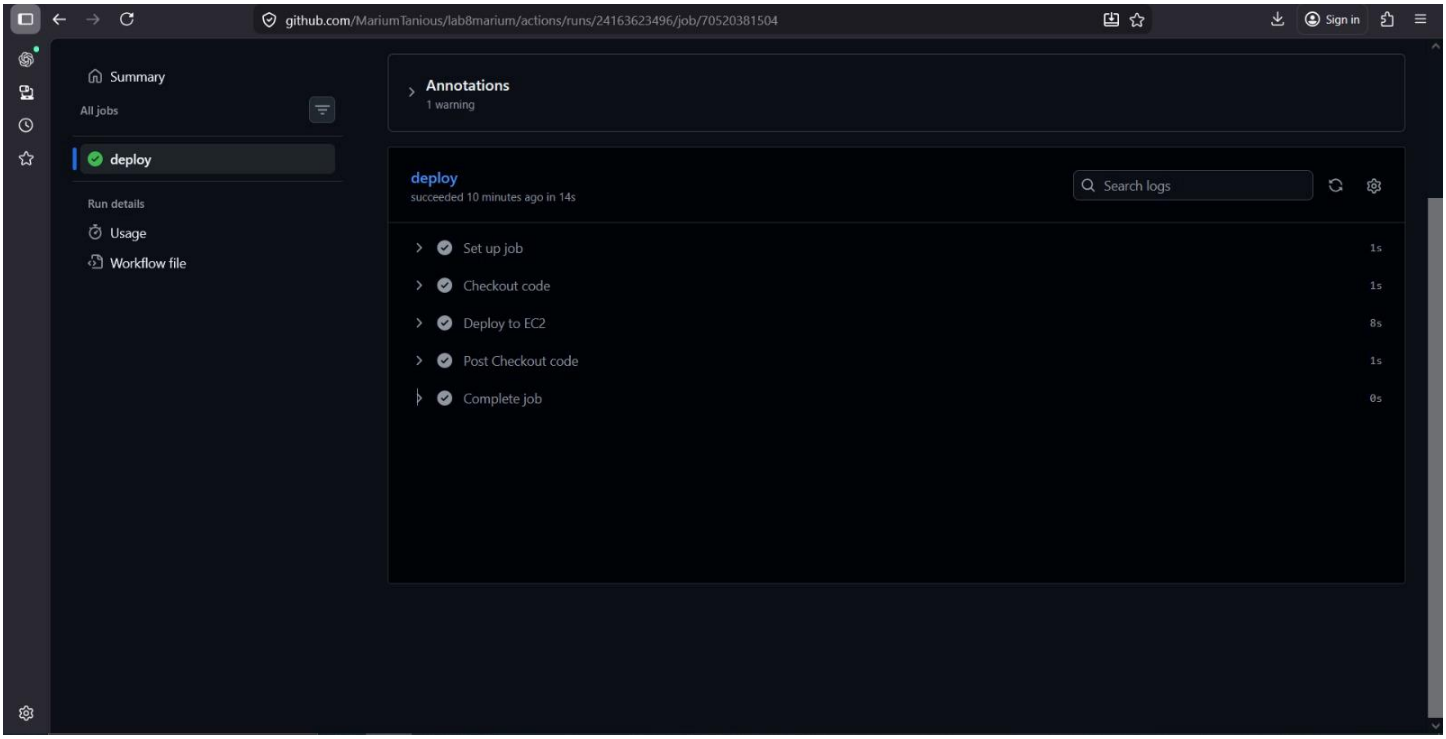
2. App Fetching Tasks:

```
ubuntu@ip-172-31-10-11:~$ cd ~/lab8marium
ubuntu@ip-172-31-10-11:~/lab8marium$ sudo docker-compose up -d --build
Creating network "lab8marium_default" with the default driver
Building app
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.941MB
Step 1/7 : FROM node:18
 ---> b50082bc3670
Step 2/7 : WORKDIR /app
 ---> Using cache
 ---> c7d649e33e0c
Step 3/7 : COPY package*.json ./
 ---> Using cache
 ---> 63ec6312bd30
Step 4/7 : RUN npm install
 ---> Using cache
 ---> 3408690607a8
Step 5/7 : COPY . .
 ---> Using cache
 ---> aa6f24aca996
Step 6/7 : EXPOSE 3000
 ---> Using cache
 ---> 68e705a40fff
Step 7/7 : CMD ["node", "app.js"]
 ---> Using cache
 ---> e36025522e94
Successfully built e36025522e94
Successfully tagged lab8marium_app:latest
Creating lab8marium_mongo_1 ... done
Creating lab8marium_app_1   ... done
ubuntu@ip-172-31-10-11:~/lab8marium$ sudo docker-compose ps
       Name                     Command               State                    Ports
-------------------------------------------------------------------------------------------------
lab8marium_app_1     docker-entrypoint.sh node  ...   Up      0.0.0.0:3000->3000/tcp,:::3000->3000/tcp
lab8marium_mongo_1   docker-entrypoint.sh mongod      Up      27017/tcp
ubuntu@ip-172-31-10-11:~/lab8marium$ curl localhost:3000/tasks
{"done":[{"id":1,"name":"Milk","status":"done"},{"id":2,"name":"Eggs","status":"done"}],"pending":[{"id":3,"name":"Bread","status":"pending"},{"id":4,"name":"Butter","status":"pending"},{"id":5,"name":"Orange juice","status":"pending"},{
"id":6,"name":"Tea","status":"pending"}]}ubuntu@ip-172-31-10-11:~/lab8marium$
```

Task 2 – Update the Pipeline to Deploy the Compose Stack

The two containers communicate internally using Docker Compose networking, which allows the app container to reach the database container by its service name without exposing the database port publicly.



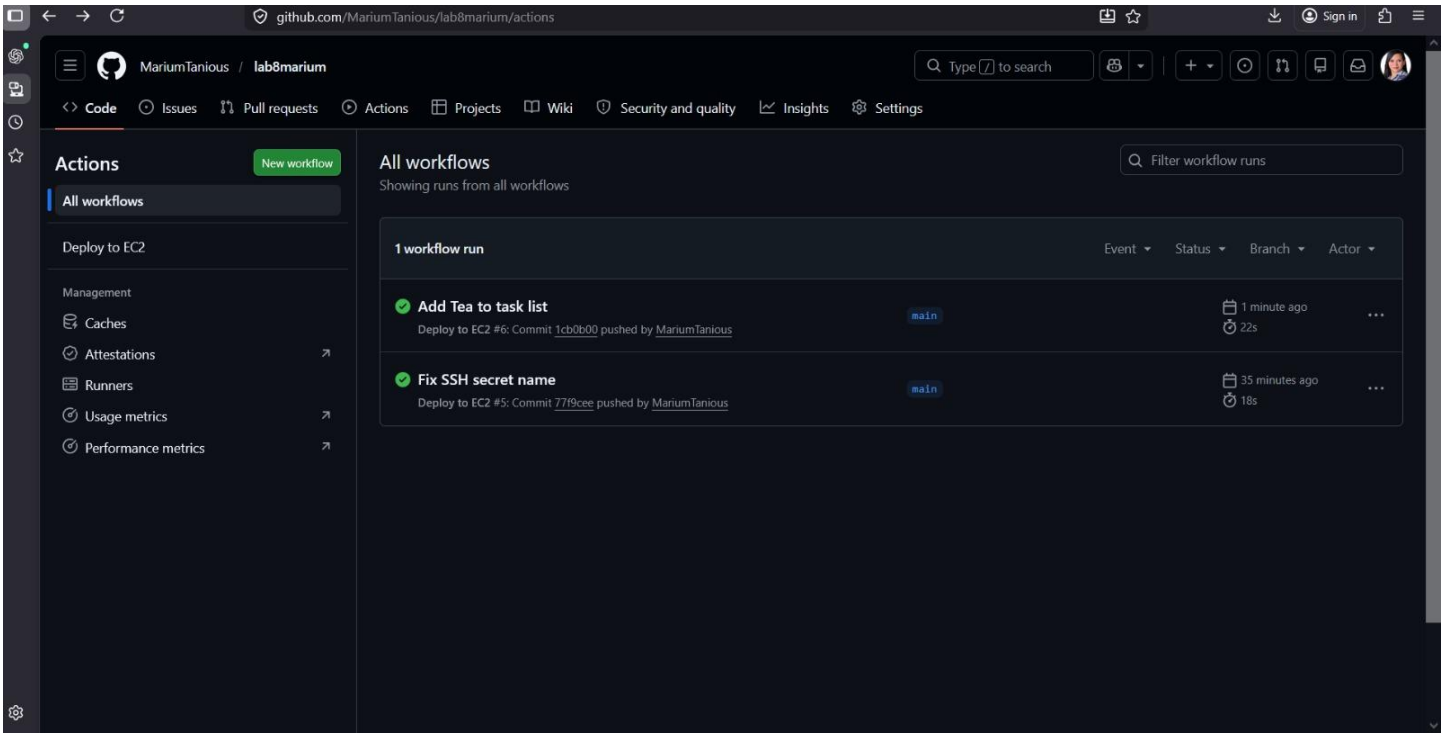


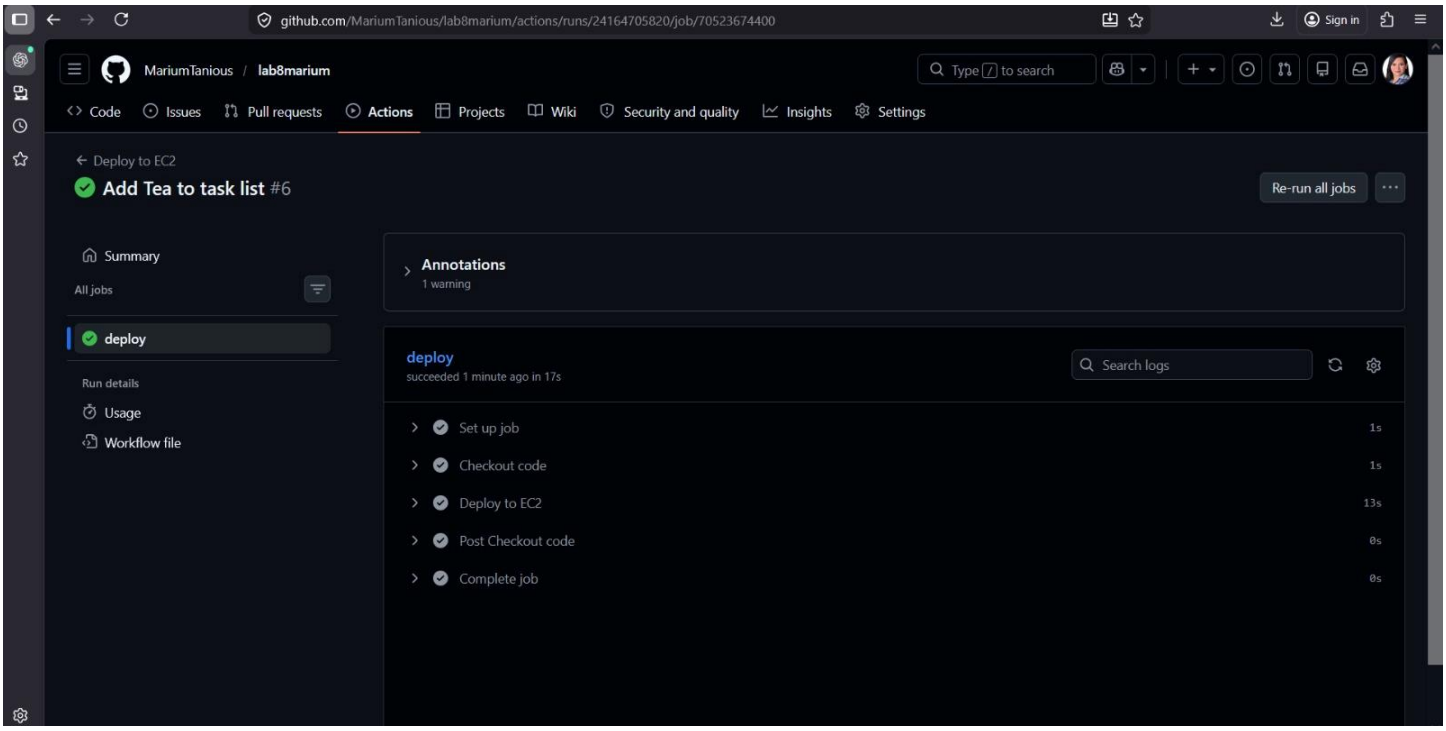
```
ubuntu@ip-172-31-10-11:~/lab8marium$ nano .github/workflows/deploy.yml
ubuntu@ip-172-31-10-11:~/lab8marium$ nano .github/workflows/deploy.yml
ubuntu@ip-172-31-10-11:~/lab8marium$ git add .github/workflows/deploy.yml
ubuntu@ip-172-31-10-11:~/lab8marium$ git commit -m "Update pipeline to deploy with Docker Compose"
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .github/workflows/deploy.yml.save
        .github/workflows/deploy.yml.save.1
        node_modules/
        package-lock.json

nothing added to commit but untracked files present (use "git add" to track)
ubuntu@ip-172-31-10-11:~/lab8marium$ git push origin main
Everything up-to-date
ubuntu@ip-172-31-10-11:~/lab8marium$ docker compose ps
docker: unknown command: docker compose

Run 'docker --help' for more information
ubuntu@ip-172-31-10-11:~/lab8marium$ docker-compose ps
NAME                 IMAGE            COMMAND                  SERVICE   CREATED         STATUS         PORTS
lab8marium-app-1     lab8marium-app   "docker-entrypoint.s…"   app       7 minutes ago   Up 7 minutes   0.0.0.0:3000->3000/tcp, :::3000->3000/tcp
lab8marium-mongo-1   mongo:6          "docker-entrypoint.s…"   mongo     7 minutes ago   Up 7 minutes   27017/tcp
ubuntu@ip-172-31-10-11:~/lab8marium$ curl http://localhost:3000/tasks
{"done":[{"id":1,"name":"Milk","status":"done"},{"id":2,"name":"Eggs","status":"done"}],"pending":[{"id":3,"name":"Bread","status":"pending"},{"id":4,"name":"Butter","status":"pending"},{"i
d":5,"name":"Orange juice","status":"pending"},{"id":6,"name":"Tea","status":"pending"}]}ubuntu@ip-172-31-10-11:~/lab8marium$
```

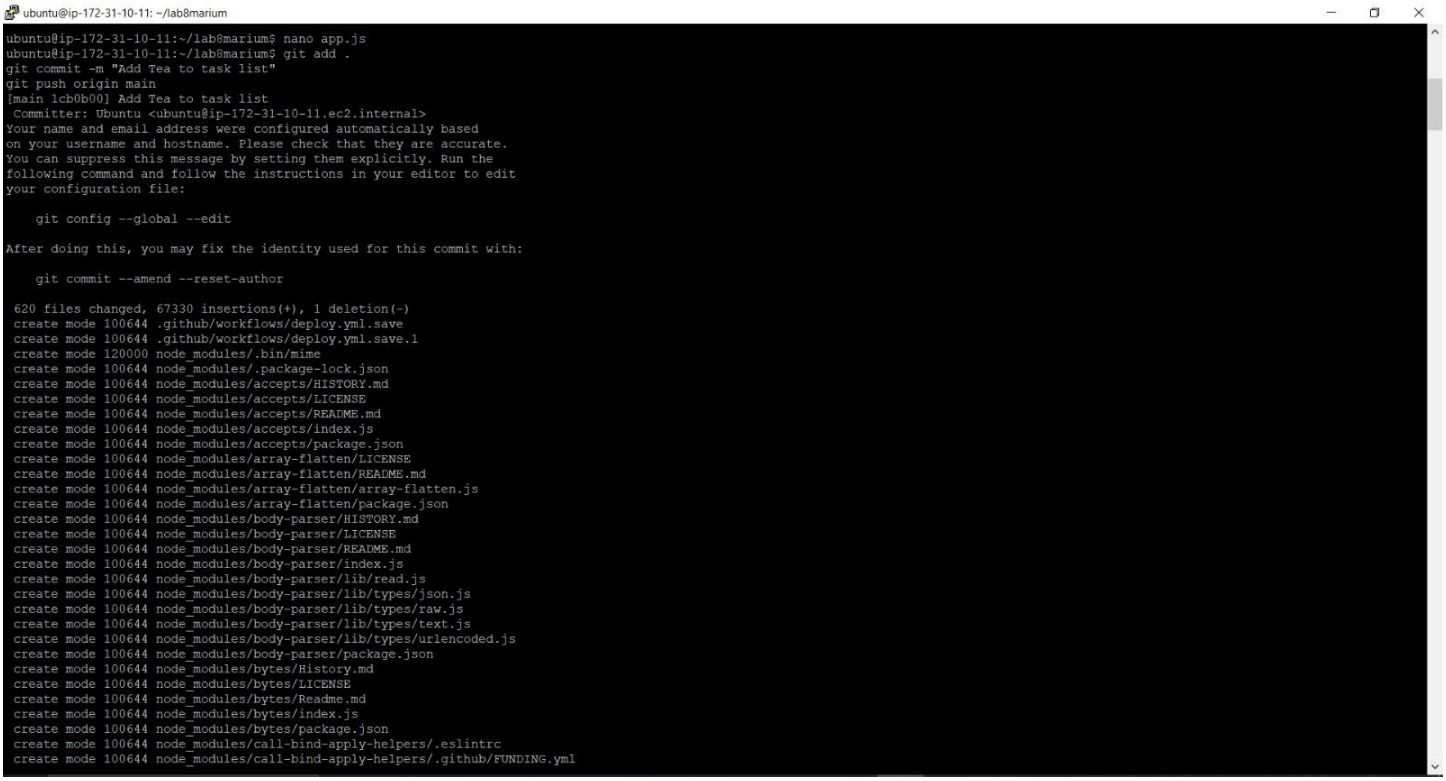
Task 3 – Trigger a Full Round-Trip Change

Docker Hub acts as the intermediary at the point when docker compose pull is executed in the pipeline.

This matters because the EC2 instance does not need to clone the GitHub repository or build images locally; it simply pulls the pre-built images from Docker Hub, ensuring updates are delivered even if direct GitHub access is restricted.







```
ubuntu@ip-172-31-10-11: ~/lab8marium
ubuntu@ip-172-31-10-11:~/lab8marium$ nano app.js
ubuntu@ip-172-31-10-11:~/lab8marium$ git add .
git commit -m "Add Tea to task list"
git push origin main
[main 1cb0b00] Add Tea to task list
 Committer: Ubuntu <ubuntu@ip-172-31-10-11.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 620 files changed, 67330 insertions(+), 1 deletion(-)
 create mode 100644 .github/workflows/deploy.yml.save
 create mode 100644 .github/workflows/deploy.yml.save.1
 create mode 120000 node_modules/.bin/mime
 create mode 100644 node_modules/.package-lock.json
 create mode 100644 node_modules/accepts/HISTORY.md
 create mode 100644 node_modules/accepts/LICENSE
 create mode 100644 node_modules/accepts/README.md
 create mode 100644 node_modules/accepts/index.js
 create mode 100644 node_modules/accepts/package.json
 create mode 100644 node_modules/array-flatten/LICENSE
 create mode 100644 node_modules/array-flatten/README.md
 create mode 100644 node_modules/array-flatten/array-flatten.js
 create mode 100644 node_modules/array-flatten/package.json
 create mode 100644 node_modules/body-parser/HISTORY.md
 create mode 100644 node_modules/body-parser/LICENSE
 create mode 100644 node_modules/body-parser/README.md
 create mode 100644 node_modules/body-parser/index.js
 create mode 100644 node_modules/body-parser/lib/read.js
 create mode 100644 node_modules/body-parser/lib/types/json.js
 create mode 100644 node_modules/body-parser/lib/types/raw.js
 create mode 100644 node_modules/body-parser/lib/types/text.js
 create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
 create mode 100644 node_modules/body-parser/package.json
 create mode 100644 node_modules/bytes/History.md
 create mode 100644 node_modules/bytes/LICENSE
 create mode 100644 node_modules/bytes/Readme.md
 create mode 100644 node_modules/bytes/index.js
 create mode 100644 node_modules/bytes/package.json
 create mode 100644 node_modules/call-bind-apply-helpers/.eslintrc
 create mode 100644 node_modules/call-bind-apply-helpers/.github/FUNDING.yml
```

ubuntu@ip-172-31-10-11: ~/lab8marium

```
 create mode 100644 node_modules/side-channel/LICENSE
 create mode 100644 node_modules/side-channel/README.md
 create mode 100644 node_modules/side-channel/index.d.ts
 create mode 100644 node_modules/side-channel/index.js
 create mode 100644 node_modules/side-channel/package.json
 create mode 100644 node_modules/side-channel/test/index.js
 create mode 100644 node_modules/side-channel/tsconfig.json
 create mode 100644 node_modules/statuses/HISTORY.md
 create mode 100644 node_modules/statuses/LICENSE
 create mode 100644 node_modules/statuses/README.md
 create mode 100644 node_modules/statuses/codes.json
 create mode 100644 node_modules/statuses/index.js
 create mode 100644 node_modules/statuses/package.json
 create mode 100644 node_modules/toidentifier/HISTORY.md
 create mode 100644 node_modules/toidentifier/LICENSE
 create mode 100644 node_modules/toidentifier/README.md
 create mode 100644 node_modules/toidentifier/index.js
 create mode 100644 node_modules/toidentifier/package.json
 create mode 100644 node_modules/type-is/HISTORY.md
 create mode 100644 node_modules/type-is/LICENSE
 create mode 100644 node_modules/type-is/README.md
 create mode 100644 node_modules/type-is/index.js
 create mode 100644 node_modules/type-is/package.json
 create mode 100644 node_modules/unpipe/HISTORY.md
 create mode 100644 node_modules/unpipe/LICENSE
 create mode 100644 node_modules/unpipe/README.md
 create mode 100644 node_modules/unpipe/index.js
 create mode 100644 node_modules/unpipe/package.json
 create mode 100644 node_modules/utils-merge/.npmignore
 create mode 100644 node_modules/utils-merge/LICENSE
 create mode 100644 node_modules/utils-merge/README.md
 create mode 100644 node_modules/utils-merge/index.js
 create mode 100644 node_modules/utils-merge/package.json
 create mode 100644 node_modules/vary/HISTORY.md
 create mode 100644 node_modules/vary/LICENSE
 create mode 100644 node_modules/vary/README.md
 create mode 100644 node_modules/vary/index.js
 create mode 100644 node_modules/vary/package.json
 create mode 100644 package-lock.json
Enumerating objects: 720, done.
Counting objects: 100% (720/720), done.
Delta compression using up to 2 threads
Compressing objects: 100% (660/660), done.
Writing objects: 100% (716/716), 716.46 KiB | 7.39 MiB/s, done.
Total 716 (delta 131), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (131/131), completed with 1 local object.
To github.com:MariumTanious/lab8marium.git
   77f9cee..1cb0b00  main -> main
ubuntu@ip-172-31-10-11:~/lab8marium$
```

```
Enumerating objects: 720, done.
Counting objects: 100% (720/720), done.
Delta compression using up to 2 threads
Compressing objects: 100% (660/660), done.
Writing objects: 100% (716/716), 716.46 KiB | 7.39 MiB/s, done.
Total 716 (delta 131), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (131/131), completed with 1 local object.
To github.com:MariumTanious/lab8marium.git
   77f9cee..1cb0b00  main -> main
ubuntu@ip-172-31-10-11:~/lab8marium$ docker-compose ps
NAME                  IMAGE             COMMAND                  SERVICE   CREATED          STATUS          PORTS
lab8marium-app-1      lab8marium-app    "docker-entrypoint.s…"   app       13 minutes ago   Up 13 minutes   0.0.0.0:3000->3000/tcp, :::3000->3000/tcp
lab8marium-mongo-1    mongo:6           "docker-entrypoint.s…"   mongo     13 minutes ago   Up 13 minutes   27017/tcp
ubuntu@ip-172-31-10-11:~/lab8marium$ curl http://localhost:3000/tasks
{"done":[{"id":1,"name":"Milk","status":"done"},{"id":2,"name":"Eggs","status":"done"}],"pending":[{"id":3,"name":"Bread","status":"pending"},{"id":4,"name":"Butter","status":"pending"},{"i
d":5,"name":"Orange juice","status":"pending"},{"id":6,"name":"Tea","status":"pending"}]}ubuntu@ip-172-31-10-11:~/lab8marium$
```